

UAVI — Uncertainty Estimation

Technique Specification for a Black-Box (Gemini API) Implementation of Paper A

VisionInspect Research Extension Program | Internal Technical Note | Document v1.0

1. Goal

Paper A (UAVI) proposes a multi-dimensional uncertainty estimation layer for the VisionInspect pipeline, replacing a single opaque model-confidence score with a calibrated probability of correctness. Because VisionInspect's multimodal analysis runs entirely against the Gemini API, the system has no access to model internals — no logits over the full vocabulary, no weight ensembles, no dropout sampling, and no gradient-based signals. This document specifies how each of the four selected techniques below reconstructs a genuine, evaluable uncertainty signal using only black-box API calls and local computation, and defines the numeric outputs each technique must report for the resulting paper to meet Q1-journal / top-tier-venue evidentiary standards.

A Q1-level treatment requires more than a single point estimate per technique: every technique below must report (a) the raw signal computed per inspection, (b) an aggregate statistic across the evaluation set, (c) a discriminative metric showing the signal actually separates correct from incorrect predictions, and (d) a comparison against a naive baseline. These four output classes are what referees expect before accepting an uncertainty-estimation claim.

2. Technique 1 — Prediction Uncertainty via Self-Consistency Sampling

The most direct black-box proxy for a model's internal uncertainty is behavioral variance: how much the model's own output changes when it is asked the same question multiple times under non-zero temperature. This technique treats disagreement across repeated Gemini calls as the primary uncertainty signal, since a genuinely confident model should give a stable answer regardless of sampling noise.

Method

- Issue N = 5 independent Gemini calls on the identical image and inspection prompt, with temperature set above 0 (e.g. 0.7–1.0) so sampling variance is actually exercised.
- From each of the 5 responses, extract the predicted defect label, severity tier, and the model's self-reported confidence score.
- Compute the empirical label distribution across the 5 samples (e.g. 4/5 "crack", 1/5 "scratch").
- Reserve this call budget for borderline cases only (single-shot confidence in a mid-range band, e.g. 0.35–0.75) to control inference cost; route obviously high- or low-confidence cases through a single call.

Formula

$$U_{\text{prediction}} = 1 - \max_c \left(\text{count}(\text{label} = c) / N \right) \quad (\text{normalized label-disagreement rate across } N=5 \text{ samples})$$

Required Numeric Outputs (Q1 bar)

Output	Definition	Q1 Target / Reporting Requirement
Label agreement rate	Fraction of the 5 samples matching the plurality label, averaged over the test set.	Report mean $\pm$ std across the full evaluation set ($n \geq 500$ inspections).
Severity agreement rate	Same computation applied to the severity tier rather than the defect label.	Reported separately from label agreement; both required.
Sampling-variance AUROC	AUROC of U_prediction as a binary predictor of "model was wrong" on held-out ground truth.	≥ 0.70 to claim the signal is informative; must exceed the single-shot self-reported confidence baseline AUROC by a stated margin.
Cost-of-sampling delta	Extra API calls and latency incurred per inspection from the 5x sampling protocol, and the fraction of cases actually routed to multi-sampling.	Report mean extra calls/inspection and mean added latency (ms); required for a fair cost-benefit claim.
Baseline comparison	Same AUROC/agreement analysis computed for raw single-call self-reported confidence.	Must be reported side-by-side in the same table as U_prediction, not in a separate section.

Referee-facing caveat: self-consistency variance is a proxy for epistemic uncertainty under sampling, not a calibrated posterior. The paper must state this limitation explicitly and must not describe U_prediction as "model confidence" without qualification.

3. Technique 2 — Image Quality Uncertainty via Classical Computer Vision

Unlike the other three techniques, image-quality uncertainty requires no calls to Gemini at all. It is computed locally and deterministically from the pixel data using standard CV operators, which makes it the cheapest and most reproducible component of the uncertainty stack, and a natural first gate before any API spend occurs.

Method

- Blur: variance of the Laplacian of the grayscale image; low variance indicates a soft or out-of-focus capture.
- Exposure: histogram analysis for clipped highlights/shadows (fraction of pixels saturated at 0 or 255) to flag over- or under-exposure.
- Resolution: compare the image's effective resolution and pixel density over the inspected region against a minimum-resolution threshold set from the defect taxonomy.
- Occlusion / glare: local contrast and specular-highlight detection to flag regions where the surface of interest is partially obscured.
- Combine the four sub-scores into a single normalized U_image score via min-max scaling against calibration-set percentiles, then a weighted sum.

Formula

$$U_{\text{image}} = w_b \cdot \text{blur_score} + w_e \cdot \text{exposure_score} + w_r \cdot \text{resolution_score} + w_o \cdot \text{occlusion_score},$$

each sub-score $\in [0, 1]$

Required Numeric Outputs (Q1 bar)

Output	Definition	Q1 Target / Reporting Requirement
Per-operator thresholds	The exact Laplacian-variance, histogram-clipping, resolution, and glare thresholds used, and how they were tuned.	Thresholds must be derived from a labeled calibration set, not hand-picked; report the tuning procedure (e.g. ROC-optimal cutoff).
Correlation with downstream error	Point-biserial correlation (or AUROC) between U_image and inspection error rate.	Report correlation coefficient with a p-value / confidence interval; expect a smaller effect size than U_prediction, but must be non-trivial (e.g. $r \geq 0.15$, $p < 0.01$).
Quality-stratified accuracy drop	Inspection accuracy for the lowest-quality quartile vs. the highest-quality quartile of images, by U_image score.	Report accuracy gap in percentage points between quartiles as the headline evidence that the signal matters operationally.
Compute cost	Wall-clock time per image for the full CV pipeline.	Report mean/percentile latency (ms) on the target deployment hardware; should be sub-100ms to justify running on every image.

4. Technique 3 — Semantic Uncertainty via Clustering of Free-Text Reasoning

This technique reuses the same 5 samples gathered for Technique 1 but looks past the discrete label to the model's stated reasoning. Two samples can agree on "crack" while citing entirely different visual evidence for it — that divergence is a distinct and often earlier warning sign of an unreliable prediction than label disagreement alone, and is the core justification for treating semantic uncertainty as a separate component rather than folding it into U_prediction.

Method

- For each of the 5 sampled responses, extract the free-text evidence/rationale span (not just the label).
- Embed each rationale with a sentence-embedding model to obtain 5 vectors per inspection.
- Compute pairwise cosine similarity across the 5 embeddings, or cluster them (e.g. agglomerative clustering with a fixed similarity threshold) and count the number of distinct semantic clusters.
- Flag "same-label, divergent-rationale" cases explicitly, since these are the failure mode this technique is specifically designed to catch and are invisible to label-only agreement metrics.

Formula

$$U_{\text{semantic}} = 1 - \text{mean}_{i \neq j} (\cos_sim(\text{embed}(\text{rationale}_i), \text{embed}(\text{rationale}_j))) \quad \text{averaged over all pairs among the 5 samples}$$

Required Numeric Outputs (Q1 bar)

Output	Definition	Q1 Target / Reporting Requirement
Mean pairwise rationale similarity	Average cosine similarity across rationale-embedding pairs, reported per inspection and aggregated over the evaluation set.	Report distribution (mean, median, IQR), not a single mean, since this signal is expected to be skewed.
Cluster count distribution	Histogram of the number of distinct rationale clusters (1 to 5) across the evaluation set.	Report full histogram; used as a human-interpretable diagnostic alongside the continuous score.
Same-label / divergent-rationale rate	Fraction of inspections where the label agreement rate is high ($\geq 4/5$) but rationale clusters ≥ 2 .	Must be reported explicitly as the key novel-failure-mode statistic distinguishing this technique from Technique 1; report its own error rate on held-out ground truth.
Incremental AUROC over label-only signal	AUROC for error prediction using [U_prediction] alone vs. [U_prediction + U_semantic] combined.	Must show a statistically significant AUROC improvement (e.g. DeLong test, $p < 0.05$) to justify semantic uncertainty as a separate paper contribution rather than a redundant signal.

5. Technique 4 — Evidence Consistency via Verifier Cross-Examination

The first three techniques measure variance in what the model says. This technique instead checks whether what the model says is actually supported by the image, using a second, independent pass. A separate verifier prompt (optionally on a cheaper model, since it does not need the full inspection capability) is given the image plus the claimed defect and evidence, and is asked to judge whether the evidence actually substantiates the claim. Where available, the verifier is also asked for bounding-box or region citations, which can be checked geometrically rather than just semantically.

Method

- Construct a verifier prompt: image + claimed defect label + claimed evidence text → structured judgment of support (supported / partially supported / unsupported) plus a scalar mismatch score.
- Run the verifier as an independent call, decoupled from the original inspection call, so it cannot simply restate the same reasoning chain.
- Where the API/prompt supports region grounding, request bounding boxes or crop coordinates from both the original inspection call and the verifier, and compute an IoU (intersection-over-union) between the two as a geometric consistency check.
- Score the overall evidence-consistency mismatch as a combination of the verifier's semantic judgment and the geometric IoU check when region citations are available.

Formula

$$U_{\text{evidence}} = \alpha \cdot (1 - \text{verifier_support_score}) + (1 - \alpha) \cdot (1 - \text{IoU}(\text{box}_{\text{inspection}}, \text{box}_{\text{verifier}}))$$

(IoU term omitted when region grounding unavailable, $\alpha = 1$)

Required Numeric Outputs (Q1 bar)

Output	Definition	Q1 Target / Reporting Requirement
Verifier agreement rate	Fraction of inspections the verifier labels "supported" vs. "partially/unsupported".	Report full 3-class distribution, not a collapsed binary rate.
Verifier-vs-ground-truth precision	Of the cases the verifier flags "unsupported", what fraction were confirmed actual inspection errors on held-out ground truth.	Precision ≥ 0.6 recommended as the bar for the verifier flag to be operationally useful rather than noisy.
Mean bounding-box IoU	Average IoU between the original call's cited region and the verifier's independently cited region, for cases where both provide region grounding.	Report mean IoU and the fraction of inspections for which grounding was available at all (grounding coverage rate).
Verifier-model agreement AUROC	AUROC of U_{evidence} as a predictor of inspection error, evaluated separately for same-model (Gemini-on-Gemini) and cross-model (cheaper verifier) configurations.	Both configurations must be reported; cross-model verification should be shown to reduce the correlated-error risk of a model checking itself, ideally via a lower false-negative rate on known adversarial/ambiguous cases.

Output	Definition	Q1 Target / Reporting Requirement
Verifier cost overhead	Extra API calls, tokens, and latency added per inspection by the verifier pass.	Report separately from Technique 1's sampling overhead so the two costs are not conflated in the paper's efficiency section.

6. Cross-Cutting Requirements for Q1-Level Evidence

Beyond the per-technique outputs above, a Q1 submission needs the following evaluated jointly across all four techniques, since referees will expect the composite uncertainty score — not just its individual components — to be the object under test.

- **Calibration set / held-out test set split:** a labeled dataset with confirmed ground-truth correctness, disjoint from any set used to fit the fusion weights (w_1 – w_4) or verifier thresholds, with the split size and composition explicitly reported.
- **Fusion model:** the composite U (weighted sum of the four components) must be fit via a documented calibration procedure (e.g. logistic or isotonic regression against ground-truth correctness), with reported coefficients, not hand-tuned weights.
- **Calibration metrics:** Expected Calibration Error (ECE), Brier score, and negative log-likelihood (NLL) for the fused calibrated probability, each reported with a baseline comparison against raw single-shot model confidence.
- **Discrimination metrics:** AUROC and AUPRC for using the fused U to detect incorrect predictions, plus a risk–coverage curve showing accuracy at varying levels of automatic-acceptance coverage.
- **Ablation table:** each of the four techniques removed one at a time from the fused score, reporting the resulting change in ECE and AUROC, to demonstrate each component's independent contribution.
- **Safety constraint:** critical-defect recall must be reported as maintained or improved relative to the deterministic-threshold baseline in every experimental condition above — this is treated as a non-negotiable gate, not a secondary metric.
- **Statistical significance:** confidence intervals (bootstrap resampling recommended, $n \geq 1000$ resamples) on every headline number, and significance tests for every technique-vs-baseline comparison.

Taken together, sections 2–5 define the mechanism for each of the four selected black-box techniques, and section 6 defines the composite-level evidence a reviewer at a top-tier venue would require before accepting that the resulting calibrated uncertainty score is both statistically sound and operationally meaningful for industrial inspection.